

# AYMANE AIT DADS

Software Engineer Intern - Kernels | AI Inference Optimization · CUDA/Triton Kernel Engineering · LLM Fine-Tuning

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## PROFESSIONAL SUMMARY

Engineering student with hands-on CUDA and Triton kernel debugging experience resolving 5 Blackwell GPU (sm\_120) incompatibilities in strict dependency order to enable 30B Mamba-Transformer MoE inference on a 95GB VRAM system. Delivered a production clustering pipeline on 100K+ fiber-optic network records at Orange Maroc, reducing manual analysis time by ~60% through automated feature engineering and model evaluation. Proficient in Python, C-level GPU debugging, PyTorch, Triton, and hardware-software interaction across CUDA compute targets - skills directly mapped to kernel development and inference stack optimization. Available for relocation to the California Bay Area; visa sponsorship required (currently based in France, eligible to apply for US internship visa).

## CORE COMPETENCIES

AI/LLM kernel inference | kernel performance profiling | C/C++ code optimization | hardware-software interaction | compute development frameworks | Triton-lang | problem-solving & debugging | AI inference stack

## WORK EXPERIENCE

### Orange Maroc

Jul 2024 - Aug 2024

#### Data Science Intern - Casablanca, Morocco

- Developed an end-to-end machine learning pipeline on **100K+ fiber-optic network records** using Random Forest and K-Means to classify network quality across thousands of nodes.
- Engineered a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering, cross-validated model evaluation, and reproducible Scikit-learn workflows.
- Delivered stakeholder presentations translating model outputs into maintenance recommendations, visualized via **Power BI** dashboards for non-technical audiences.

## PROJECTS

### NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard (0.80/1.0)** in a \$106K+ prize-pool competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) via SFTTrainer on 7,828 chain-of-thought examples.
- Resolved **5 Blackwell GPU (sm\_120) incompatibilities** in strict dependency order - including monkey-patched caching allocator, stubbed mamba3/cutlass imports, disabled Mamba CUDA fast-path, fixed Triton ptxas permissions, and replaced rmsnorm\_fn with a pure PyTorch fallback - then ran 20+ single-variable ablations confirming completion-only loss and packing configurations.

PyTorch · Triton · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Mamba

### AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in EURECOM Class

- Ranked **1st in EURECOM class leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and unfreezing up to 8 transformer blocks across 250K training images from 25+ generator types.
- Submitted to **NTIRE 2026 @ CVPR CodaBench** international benchmark; ablation studies confirmed 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC) due to out-of-distribution generalization, validated by a 3-model ensemble with 10-view TTA.

PyTorch · OpenCLIP · ViT · FP16 mixed precision · Kaggle T4 GPU

## EDUCATION

### Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Cloud Computing, Image Security, Statistics, Web Applications

### CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

## SKILLS

**Kernel & Systems Engineering:** Python | C++ (code optimization) | CUDA debugging | Triton-lang | PyTorch | Bash | Linux

**ML Frameworks & LLM Engineering:** Hugging Face Transformers | Unsloth | TRL | PEFT | LoRA / QLoRA | vLLM | Scikit-learn | Mamba-Transformer MoE

**Data, Profiling & MLOps:** Pandas | NumPy | FP16/BF16 mixed precision | ablation studies | feature engineering | Power BI | Git